Andrew Pols

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Education

University of Toronto
Expected Graduation: May 2028

- Bachelor of Science in **Computer Science and Statistics** **GPA: 4.00 / 4.00**
- **Coursework**: Software Design, Computer Organization, Data Science, Software Tools and Systems Programming
- **Awards**: Dean's List (4 terms), U of T Scholar, Dorothy Helen McRobb Scholarship, Governor General's Bronze Medal

Experience

IBM
May 2026 – May 2027

Software Developer Intern

- Shipped a dependency graph feature in React and TypeScript for Planning Analytics Workspace, which **Fortune 500** clients had asked for and competitors already offered, **automating relationship parsing** across models with thousands of objects.
- Reduced dependency tracing from **hours or days** of manual work to **seconds**, scoping the workflow with design, PM, and QA.
- Cut initial load on a **2,000-object** model from 30s and 50 requests to **3s and 1 request** by adding lazy-loading, request aggregation, and caching to a recursive TM1 REST API traversal; each later dependency loads in **under 500ms**.
- Demoed to IBM leadership, who presented it with PM at the **TM1 Anwendertag** and **IBM TechXChange** conferences.
- Integrating LLM-based localization into text widgets so clients see saved text in any of PAW's supported languages.

Projects

ScholarSearch: Academic Search Engine | *GitHub* Python | FastAPI | PostgreSQL | ChromaDB | Redis | React

- Engineered a hybrid BM25 and semantic search engine over 1M DBLP papers with FastAPI and ChromaDB, serving cold queries in a **334ms** median (p99 under 600ms) and cached queries in **1.2ms**.
- Embedded all 1M papers at **313 papers/sec**, then profiled the query slowdown to a 300ms in-memory BM25 scan.
- Visualized citations with a D3.js graph in React that shows how the papers behind a search cite each other.

Multiplayer Pokémon Battle Server | *GitHub* C | TCP Sockets | UNIX | ncurses

- Architected a Pokémon battle server in C that handles **1,000 concurrent TCP clients** in a single-threaded select() loop.
- Sustained a median **51k messages/sec** at **26ms p99** on loopback across 10 runs with zero connection or sanitizer errors.
- Defined a binary protocol with a 5-byte header and buffered reads so fragmented TCP packets parse correctly.
- Implemented 1v1 matchmaking with a lobby, rematches, forfeits, live chat, disconnect handling, and an ncurses client.

SudokuCV: Computer Vision Sudoku Solver | *GitHub* Java | OpenCV | Tesseract OCR

- Built a Java app that solves Sudoku from an image or webcam feed using OpenCV, Tesseract OCR, and backtracking.
- Detected the grid in 2ms per frame and solved **27 of 27** test puzzles with a 0.5ms median solve time.
- Raised exact grid reads from 7% to **92%** and digit accuracy from 89% to **99.96%** across 789 phone-video frames of 3 printed puzzles by fixing Hough-based line removal that erased stacked digits.

FindMySound: Music Recommender | *GitHub* Python | Django | SQLite | Scikit-Learn | React

- Developed a web app that recommends music from a user's Spotify library using the Spotify Web API and OAuth 2.0.
- Ranked songs by cosine similarity on BERT feature vectors, reaching **95% satisfaction** across 5,000 songs for 50 users.
- Cached each song's audio features after a **2s** first analysis, and ranked **400 candidates** against a 100-song playlist in **0.2ms**.
- Designed the Django backend with a custom user model, JWT authentication, SQLite, and REST API endpoints.

Technical Skills

- **Programming Languages**: Python, TypeScript, JavaScript, Java, C, Go, SQL, HTML5, CSS
- **Frameworks & Libraries**: React, Node.js, FastAPI, Django, Flask, Next.js, Scikit-learn, OpenCV, REST, GraphQL
- **Databases & Cloud**: PostgreSQL, MySQL, MongoDB, Redis, ChromaDB; AWS, Azure
- **Developer Tools**: Git, Docker, Vite, Jenkins, UNIX, Make, Valgrind, AddressSanitizer, Jira